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RE-IMAGINING THE LANDSCAPE OF FUTURE FARMING

MARTIN ABILDGAARD PADALAK
IT UNIVERSITY OF COPENHAGEN
ABPA@ITU.DK

TOM JENKINS
IT UNIVERSITY OF COPENHAGEN
TOMJE@ITU.DK

ABSTRACT

This paper explores how participatory design methods can engage future farmers in imagining sustainable agricultural futures in Denmark. Denmark's agricultural sector faces a "crisis of imagination" where dominant paradigms in farming are hard to challenge. Building on Ingold's notion of the *landscape* as a relational, temporal space for *dwelling*, we propose a conceptualization of the landscape game that enables farming students to articulate their visions of future farming. The design game creates opportunities for participants to situate their imagined farms within broader social, environmental, and technical contexts while exploring how these might evolve over time. This approach aims to generate new agricultural imaginaries that move beyond current techno-solutionist or radical transformation narratives, while supporting an underrepresented political public in developing their own perspective on contested agricultural futures that are both speculative and grounded in lived experience.

INTRODUCTION

Denmark is a country with a long and proud history of farming. The cultural effect of Danish agriculture has been praised in public discourse as a driving force of the country's progress, especially by right-of-center politicians. It is a sector that has reshaped terrain, culture, and language as well as left a legacy in the many Danish surnames derived from farms. Morten Korch's novels from the early 20th century offered Danes an idyllic image of rural life. Many of his stories were later adapted for the screen (Figure 1), becoming widely popular and a prototypical image of agrarian bliss. However, the writer and journalist Rasmus Haage Dalland has argued that Korch's idyllic images were misleading even in their own time. They presented an already-ended era of independent smallhold farmers and family patriarchs, while in reality their real-world counterparts were already fighting a losing battle to industrialization (2023).



Figure 1: Film poster for "Fætrene på Torndal" based on the novel "Kampen om Torndal" by Korch (image via the Danish Film Institute).

In 1960, the agricultural sector employed 300,000 full-time workers and only 34,471 in 2020 (Danmarks Statistik, 1961, 2022). Similarly, from the 1960s' count of close to 200,000 farms, only 6,294 remained in 2023 (Kyed, 2000; Danmarks Statistik, 2024). At the same time, farms became fewer and larger, and the workforce has shrunk, while Danish livestock production has grown to produce more than 200 million animals annually (Lundsgaard et al., 2022)—releasing one fourth of the total greenhouse gas emissions of Denmark. Danish agricultural land also takes up nearly 60% of the country's area, but cultivating just the imported soy used as fodder for these animals requires an area corresponding to a further fourth of that (ibid.). Nitrogen leaching caused by excessive spreading of slurry has led to most Danish waters being affected by oxygen depletion. In September 2024, these affected areas exceeded the size of Funen and Zealand put together (Hansen, and Rytter, 2024). Danish agricultural practices are not only leaching to the sea. For the fifth year in a row, pesticide residue was found in more than half of the groundwater wells used by waterworks (Brüsch, 2025: GEUS, 2025).

Despite its negative impact on the environment, the sector's economic significance is marginal (Nørtoft, 2018). If the cost of restoring the damage already done

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was accounted for, the entire sector would be a huge economic loss (Sloth, 2021). Still, Denmark's political parties tend to avoid any legislation that is undesired by the industry's main actors. The Danish government and people know that their food is causing crisis, and so "tripartite negotiations" between the agriculture, industry, politicians, environmental advocates and municipalities–modelled after the structure of the labor market—have recently concluded in an agreement that has predictably disappointed everyone. This resulted in voluntary targets that are both not very specific about how goals should be met, as well as not very ambitious in terms of change more fundamentally, choosing a carbon tax measure effectively lower than the minimum recommendations made by an earlier governmental analysis (Gammelgård-Larsen, 2024).

Den Grønne Ungdomsbevægelse (DGUB, or the Green Youth Movement), decried the results of political decision-making like the above: "*We believe that we are in the very middle of a 'forestillingskrise' (crisis of imagination), and that it takes great new dreams for us to begin to faintly see the contours of a future for the agricultural sector within the planetary, economically and socially sustainable boundaries*" (Danker et al., 2024, p. 38).

The status quo for agriculture in Denmark is clearly unsustainable. Inside this complex political contest, there are huge amounts of financial investment—and perhaps a lack of voices that might offer different perspectives and orientations to a slow-motion crisis. One underrepresented voice is those of farmers themselves, particularly prospective farmers that will be operating the farms that will navigate the effects of the green transition in Denmark. Political conversations usually revolve around larger industrial concerns rather than perspectives from individuals, while drawing on the popular public image of the farmer to claim both sympathy and cultural resonance. This paper asks, how can design research engage and support new forms of participation in complex social issues of the future? What forms of engagement can help support imagining new worlds amid a crisis of imagination?

BACKGROUND

Topics related to food systems have been receiving increased interest from design research scholars who inquire into the particulars of food, including among others human/soil relations (Lindström and Ståhl, 2019), speculations into multi-species care (Fjalland and Samson, 2019), foraging (DiSalvo and Jenkins, 2017), edible data (Wilde and Karyda, 2024) and communal dining (Wilde et al., 2021). Considering design's ancestry in handicrafts and the bond between participatory design (PD) and design research, approaching these topics via food as a design object in itself or through community engagements is a natural progression but ensures that larger-scale production—i.e. agriculture—remains underexplored.

Agriculture has always been augmented by technology, so it may not be surprising that it has received strong attention in HCI and interaction design research (Doggett et al., 2023). At the present moment, digital agriculture (DA) has become seemingly inevitable in food production research (ETC Group, 2022). These more technical disciplines' interest reflects a growing interest in agriculture as a research topic, though relatively little directly engages with the transition away from unsustainable food production. In 2023, Doggett et al. conducted a literature review of HCI research on agriculture. They found that for the most part, ICTD research projects related to agriculture concentrate on the Global South, with a top-down approach framed around a "need to address poverty at the individual and national level as well as to respond to the demands of a rising population" (ibid., p. 4). When projects do represent the Global North, they tend to focus on urban environments and are often framed around "urban population growth, the need for climate action, and the need to collectively imagine what the future of food systems should look like in urban places" (ibid., p. 4), conveniently excluding agriculture at scale.

Timothy Morton has described contemporary food production as *agrilogistics* (2016). The idea of agrilogistics emphasize the fallout of contemporary food-systems, the paradox of relying on the Earth while also exploiting it and distorting natural eco-systems so they cater to hyperoptimized man-made ones instead, and Denmark is certainly no exception to this idea. Biggs et al. used Morton's framework to analyze smallholder farmers' practices to offer insights in support of responsible designs of sustainable agriculture computing systems (2021). While both the work and practices studies are commendable, a worry might be that taking a point of departure in niche farming practices—vastly dissimilar to the agrilogistic forms of production—can make it harder to find footing to support a wider green transition of agriculture.

DESIGN RESEARCH AND FUTURES

This project uses design to investigate possible agricultural futures. By its nature, design is a discipline invested in the future (Dunne & Raby, 2013). It seeks to make new conditions by creating new materials and arrangements of things. Many classic design tools are rooted in seeking out and articulating what hasn't yet happened, such as sketches (Buxton, 2007), storyboards (Truong et al, 2006), and especially scenarios (Carroll, 2003) to do the work of capturing and conveying the future as it is to be built. Even prototypes, while made and in the world are provisional, meant to embody and represent that which could be made in the future (Warfel, 2009). The future is being brought into design

more theoretically as well. As a research discipline, futures and futuring offer a language and framework for describing and generating different futures based in a business idea of *foresight*. For example, Candy has described Dator's four "generic futures" of *continue, collapse, discipline, and transform,* as broadly representing the stories that we tell about the future, offering categories for both thinking concretely about what ideas of the future do, as well as projecting into the future in more subtle and varied ways than as "utopia" or "dystopia" (Candy, 2010: Dator, 2009). Ideas from foresight like the futures cone (where increasingly narrow ranges of possible, plausible, and probable futures are inflected by preferable futures) have been imported into design via Dunne and Raby's book *Speculative Everything* (2013), and have become the basis of a wide range of methods that critique and expand on how we imagine futures and, critically, who gets to do that imagining (Howell et al., 2021).

Design futuring approaches for the most part do not offer solutions or targets but rather aim to open critique, discussion, debate, and questioning for alternative futures, "leverag[ing] design to explore futures as a means to comment on—and potentially change—the present" (Kozubaev et al., 2020, p. 2). Here, the future is an object of discussion that is meant to both cohere political concerns in the present as well as leave the future open to be imagined in new, speculative ways that operate with fewer constraints (Søndergaard et al., 2023). The future is not bound by the present, although it does come from it, and so developing new narratives and new means of futuring might help to engage and allay crises of imagination in times of crisis.

FUTURING FOR SUSTAINABILITY

Soden et al. found that speculative design approaches (Auger, 2013; Dunne & Raby, 2013) have gained traction within the sub-genre of sustainable HCI (2021). While they see benefits like questioning values and ethics embedded in current practices and promoting public discourse regarding climate change, sustainability and the future, their analysis did not leave them encouraged. They found that speculative design most often was used to present positive visions of the future, to frame a problem and rarely to inspire new narratives that can lead to new ways of living. The authors argue that to meet the needs of the moment requires speculative design "grounded in a positive project that has some notion of how society transitions to sustainability and a sense of urgency about this process" (ibid., p. 195), agreeing with critics who call for meaningful reorientations of speculative design to meet their critical potential (ibid.). DiSalvo has discussed the use of "spectacle" and "tropes" in speculative design (2012), describing *spectacle* as extraordinary images that draw in audiences, and tropes as "tools for crafting meaning", that "enable the creative manipulation of material and form so that we can make one thing mean another thing" (ibid., p. 117). He argues that "[n]ot all speculative design must be political, but to not address politics in social contexts where they are usually present is a striking omission" (ibid., p. 118). In line with Soden et. al, he concludes that, "[t]oo often, speculative design is spectacle alone" (ibid., p. 120), lacking the content and grounding to make productive critical statements or spark public debate.

In Scandinavia, public discourse has been engaged in what the future should bring, emphasizing how to make industries and infrastructure more sustainable, e.g. via Denmark's Klimafolkemødet (The People's Climate Meeting). These concerns are reflected in the Nordic design research community who conduct research to co-imagine citizens' lives after the societal change needed to address climate and ecological crises. For example, Lindström et al. combine speculative and PD approaches to engage with a post-carbon future of the Swedish government. This work resulted in a series of painting workshops that made space for participants to 'anticipate', share and come to terms with potential losses in post-carbon futures (2021). The same authors continued exploring post-carbon futures in a series of study circles inviting citizens to "places where different futures can be sensed" (Lindström et al., 2024). These invited external specialists to share their knowledge of places and their entanglements and then had participants perform material explorations to both support and reorient participants' views of the past, present and future, initiating difficult, yet necessary, conversations on futures between strangers, anchored in everyday objects and practices.



Figure 2: Skulls showing the development from wild boar to the domestic pig and finally the modern pig (image by M.A. Padalak).

THE ART OF THE POSSIBLE

Individual farmers are bound by legislation and shaped by large industry actors' main interest to "make more from less" (L&F, 2019). The results of this century-long industrial project were put on display at the former Royal Veterinary and Agricultural School of Copenhagen in the exhibition *Production Animals: a Gallery for the Art of the Possible*.

Curated by anthropologist Nathalia Brichet, it assembled current and historical artefacts demonstrating this process of optimization: "*Scientific research conducted by the school's pioneers has contributed to the fact that Danish dairy cows now produce an average of approximately 30 liters of milk a day. And to the fact that slaughter pigs today (Figure 2) can grow from 1kg to 100kg in 5-6 months while consuming relatively low amounts of antibiotics"* (ku.dk, 2024).

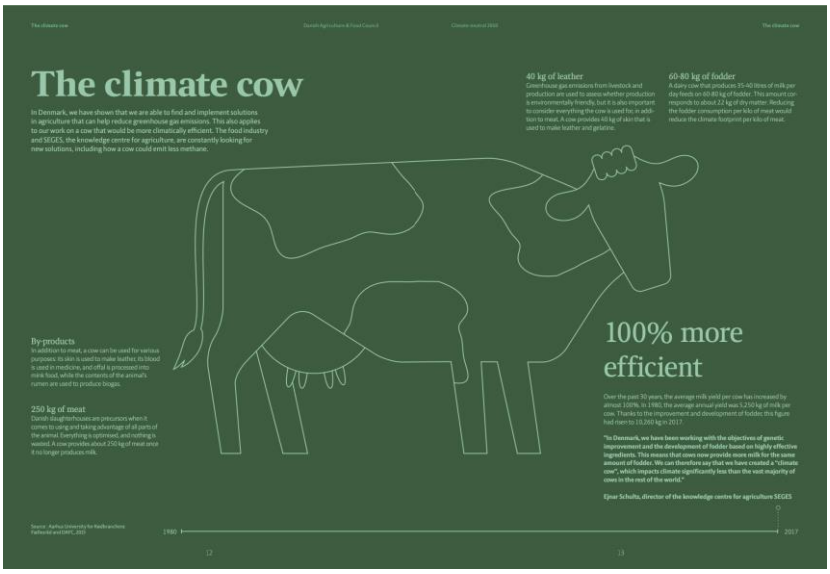


Figure 3: Spread from "Climate-neutral 2050" (L&F, 2019, pp. 12-13).

Viewed together, the artefacts make clear the Danish agricultural industry's approach to problem-solving has been to manage symptoms and not remedy the underlying causes that emerge as by-products of gaining higher "efficiency." Yet even if a few large industry actors lead the way in carrying forward a problematic mindset—in willful ignorance of its consequences—blaming them alone oversimplifies a wicked problem. It is a phenomenon that can been described as the "agricultural treadmill" (Cochrane, 1958), where improved technology increases productivity, which in turn leads to falling prices, necessitating more technology to further increase productivity to maintain profit, and so on. Farmers that do manage to keep up, aided by new technologies, expansion, and cost cutting, bind themselves to the ensuing unsustainable production forms and practices, financed by debt. Further, as the domestic food-sector is interdependent on global trade, local initiatives cannot negate this treadmill's adverse effects (Otte Hansen, 2016).

TODAY'S FUTURE FARMING IMAGINARIES

An agricultural sector that has tied itself into knots through ever-increasing efficiency leads to questions of how any transition to better alternatives might be possible. Inevitably, the visions of change have been primarily technological, "sociotechnical imaginaries" (Jasanoff and Kim, 2015) in western cultures that take as a given progress leads to better living in the future. "Farms of tomorrow" have been envisioned as current agricultural practices dropped into familiar contexts: vertical-farms and roof-top gardens in metropolitan cities, or revitalized suburban agri-neighborhoods (Doggett et al., 2023). These visions are often exciting, showcasing scenarios of high-tech harmony, back-to-basics farming approaches, or the best of both. The concepts also tend to over-promise and under-deliver, steering away from grappling with the issues at hand (Tuomisto, 2019). To illustrate the range of political arguments at hand, the following sub-sections introduce two lenses that represent opposing views of how we might create a greener agricultural sector.

CLIMATE-NEUTRAL 2050

Landbrug & Fødevarer (L&F, the Danish Agriculture & Food Council), is the Danish food industry's main trade group, with representation on 114 governmental committees, panels, and the like. By comparison, the Danish Society for Nature Conservation has seats on only 53 (FEM, 2024). L&F speaks on behalf of the entire agricultural sector, but in the organization livestock farmers, especially pig and cattle producers, assume the most central positions. As a result, L&F mainly pursues legislation and plans for farming that favors livestock production (ibid.).

"Climate-neutral 2050," was published by L&F in 2019 as a vision program to outline the Danish food industry's approach to the green transition. The ideas in it can be described as suite of 'moon-shot' technologies including a methane vaccine, supposed to make cows produce antibodies against methane producing microbes, and technology "transforming the CO_2 in the air into biochar" (ibid., p. 23) for soil improvement. This green transition is hinged on unproven concepts while also necessitating renewable energy to power energy-intensive future farms. The issues we know from current food production practices, such as the impacts of industrialization, dependence on global trade, lack of resilience from mainly monocultural production, or large-scale agriculture's effect on social well-being and job security are omitted entirely. What is carried forward is the mantra of making more from less, implying that increased efficiency inevitably leads to sustainability.

Their 100% more efficient "climate cow" (Figure 3), illustrates this conviction, highlighting mainly that the dairy industry has managed to constantly increase milk yields and that the cow's by-products "can be used for various purposes" (ibid., p. 12). *How* to realize the exact solutions, both those described and those yet to be

imagined, remains unclear. Despite its ambiguity, the presented approach has paralleled political action since the launch of *Climate-neutral 2050*.

LEVEDYGTIGT LANDBRUG

An opposing vision of agricultural futures comes from Den Grønne Ungdomsbevægelse. Founded in the spring of 2018, their goal is to "create political action at all levels by actively fighting for systemic change and sustainable alternatives to a society and way of life that occurs to the detriment of the environment, climate and people" (DGUB, 2022, p. 2). The movement's early work focused especially on activist happenings, contributions to discourse and rallying the support of the Danish public. Partly due to their work, the green transition topped the political agenda in Denmark's 2019 election—later known as "Klimavalget" (the climate vote)—where climate issues shifted more voters than seen in decades (Andersen and Hansen, 2021).

In 2024, DGUB published the farming manifesto, "Levedygigt Landbrug" (Viable Agriculture), marking a shift in their approach to instigating change. Its foreword describes having failed as an activist organization to convey earnest appreciation and sympathy for those having to transition away from everything they know and have learnt (ibid.). The manifesto signifies their attempt to re-enter the sustainable agriculture discourse "this time, humbly respecting the farmers' unfair conditions of transition, and brutal honesty regarding how difficult it will be to solve the ecological, social and economic crisis that we're in the middle of, and not least, with a holistic suggestion of an alternative to the present" (Danker et al., 2024, p. 6).

Prefaced by a summary of major issues pertaining to the Danish food industry, *Levedygtigt Landbrug* was inspired by Ernest Callenbach's 1975 science-fiction novel *Ecotopia* and the regenerative farming movement, whose fundamental idea is to sustain ourselves and other lifeforms by restoring soil fertility (ibid.). The book presents visions of sustainable futures juxtaposed with vignettes from current regenerative farms to help to tell a story of a different kind of Danish agriculture. In the visions, the late 2020s are well in the past. Buses are electric, agrarian reform has left its mark on the landscape, rural towns, and Danes' quality of life, and the drastic systemic changes have had time to mature. A "green service" lets interested youths experience way of life on a farm and the importance of cultivating the land with great care for it as well as the many beings with whom we share it. The manifesto concludes by listing concrete political actions that could lead to the future DGUB describes.

IRRECONCILABLE IMAGINARIES

Climate-neutral 2050's 24 pages feature mostly vector drawings, shades of green and limited text. *Levedygigt Landbrug,* by contrast, spans 154 pages of prose and a few monochrome images. But the imaginaries differ in more than format and length, they also represent opposite ends of a spectrum in terms of futuring. After Candy's four generic futures (Dator, 2009; Candy, 2010), the former arguably plots a course for a trajectory of *continuation* whereas the latter presents a future following that of *transformation.*

These documents illustrate two perspectives on ongoing debate in Danish society about the future of agriculture. Implicitly, they also represent a crisis of not only imagination but also of imagining—who gets to have a say and whose opinion counts in making these imaginaries. One voice is large, corporate, and reflects the industrial interests of farming in society, that are already dominant in political discussion. Conversely, the other is small, niche, and socially radical, meaning that this perspective is relatively easy to simply ignore. These ideas are certainly competing, but the power imbalance means there is no actual competition.

THE MISSING FUTURE FARMERS

We feel that the broader political conversation omits the voices of average farmers (Figure 4). While it might be thought that their perspectives are represented by the trade groups, the structural issues that have led to a dwindling number of farmers make that less clear. To develop ideas and voices of farmers, we plan to conduct workshops with students at farm schools around Denmark. Established farmers are constrained by the economic realities of the farms they operate, meaning that there may not be room for imagining alternatives. Most farm students also know people tied to the agricultural sector or have close family or friends in farming (Danske Landbrugsskoler, 2023), giving them insight and knowledge of the domain.



Figure 4: Sketch of the missing farmer (image by M.A. Padalak).

To speculate with future farmers, we find it particularly important that materials allow room for them to determine and define the relevant infrastructure and relations at play. As Rubambiza et al. have discussed, assumptions about agricultural technology are based on researchers' own experiences, often in stark contrast with the realities of rural infrastructure. Further, these perspectives discredit the historical role of farmers as producers of infrastructure and their contemporary role as tinkerers and adapters of technology for their own needs (2022). Relatedly, Hardy et al. pointed to urban spaces' domination of HCI research in 2019, arguing that it leads to rural residents being overlooked, and that attention to rural communities should "broaden

perspectives on who we design for and what practices are worth designing for" (ibid., p. 38). We see future farmers as particularly relevant participants for participatory design workshops because they operate in the middle of large forces—they have experience with farming but are not (yet) active in large industrial conglomerates or parts of student activist movements but are invested in the future of the sector that are not necessarily being accounted for in design research.

PARTICIPATORY DESIGN GAMES

Conceptual maps, like diagrams and gameboards, or metaphorical objects have worked well across a variety of PD engagements. Whether they visualize geographic locations, fictional lands or non-places, like systems, processes or power relations, maps are always conceptual renderings of what they depict. For our purposes of imagining agricultural futures, maps become relevant by offering a bird's-eye-view of the landscape.

The Landscape Game (Brandt, 2006; Halse et al., 2010) is a design game where the task is to "build a (future) landscape for the person highlighting physical surroundings or elements that augment various activities in the person's everyday life" (Brandt and Messeter, 2004, p. 125). As a research instrument, the game elicits personal values and goals in a context, making it clear what a participant finds important, but also what other players of the game value, as the game is a negotiation of interests in response to conditions established while playing (ibid.). Like many PD workshops, this game was developed to support collaboration between different stakeholders in co-design situations. Other forms of design games make the development of the game itself into a collaborative process, rather than presenting stakeholders with a pre-designed game (Eriksen and Reimer, 2015). For example, the design game UrbanTransition makes "assemblies of humans and non-humans emerge here and now in the situation," where "situations are simultaneously overflowed with relations and attachments that are connected to other times and other places" (Eriksen et al., 2014, p. 106: Latour 2004). It produces "agonistic spaces" (DiSalvo 2022; Björgvinsson, Ehn, and Hillgren 2010; Hernberg and Mazé 2018; Mouffe 2013) that support contestation and discussion about desired futures.

Particularly, map-based design games have especially been taken up by other researchers to engage with different kinds of communities to understand how those communities operate (Botero and Saad-Sulonen, 2010; Helgason et al., 2015; Jenkins, 2021; Custers et. al, 2021), allowing residents to define and adapt its components as well as expressing what they find valuable in their day-to-day context. For our purposes, we are not so much concerned with exploring participants' mutual power relations as we are with making spaces for participants to express their own values, visions, and goals and exploring them with one another. To do so, we draw on visual tropes from Danish history and culture to invite future farmers to speculate into farming futures via a map-based design game based on the medium of farming: the landscape.

THE *LANDSCAPE* AS A FRAME FOR AGRICULTURE FUTURING

We argue for developing exploratory design games that allow participants to situate and develop desirable imaginaries. Building on Tim Ingold's essay *The Temporality of the Landscape* (2002), we propose a landscape game inspired by Ingold to elicit future farmers' stories of how their future farms might relate, affect and respond to other social, environmental and technical components of the broader ecological landscape—creating new imaginaries for future agriculture.

Ingold uses the idea of the *landscape* to challenge the distinction we usually make of land and space as something that is simultaneously natural and external to humans as well as something that can be counted and accounted for. "Where land is quantitative and homogenous, the landscape is qualitative and heterogenous" (ibid., p. 190). It is something that is built over time, and whose interpretation emerges based on the history embedded in it— "an environment that itself is pregnant with the past" (ibid., p.189). The landscape is comprised of many entities—living and non-living, natural and artificial. In the landscape, there are no boundaries, just form and relation, where bodies and the landscape are figure and ground. The landscape helps to tell stories of dwelling through the idea of a *taskscape*. This is the sum of all activities which constitutes the social aspect of the landscape. Tasks, to Ingold, are the practical operations done by dwellers as part of their normal business of life. As they live, they affect the landscape and are at the same time affected by it. In this way, the landscape is the embodiment of the taskscape, as dwellers' activities collapse into features in the landscape, the world being perpetually built and rebuilt.

Ingold's perspectives are, in many ways, at odds with normative ideas of future farming. Conventional farming imaginaries in Danish culture tend to follow paradigms of numbers, boundaries and delimited views on problems and their solutions, reflected in the frequent development of techno-fixes that assume technological progress can solve the issues at hand. Motivated by eventual presumed functionality, the search for solutions often ignores both how the solutions are manufactured and how they will affect the broader context where they are implemented—a techno-solutionist ends justifying the means. While we do not intend to use this paper as counterargument to techno-

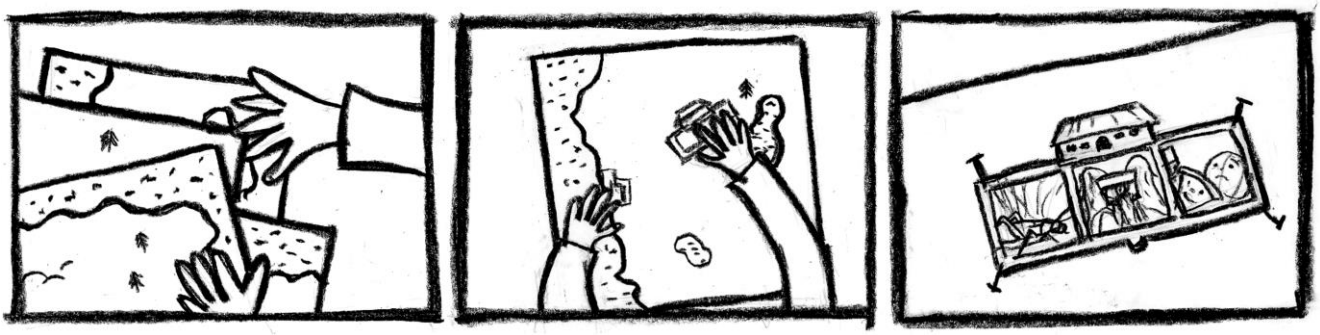


Figure 5: Three sketches of proposed activities. Left) The hands of a participant looks through their own set of gameboard maps with different topographical compositions. Middle) The participant has chosen a landscape where land meets sea, and their hands now test where to fix the game pieces - one representing their future farm, the other a metropolitan city because they'd like visits from friends in the city and hope to sell products there. Right) A farm game-piece has been fixed to the gameboard with three illustrations they liked. When the next participant is prompted to use their own selection to describe their future farm, they don't know how exactly the images they selected will relate to their own future farm. Hopefully, they'll make something up in time. (image by M.A. Padalak)

solutionism, but rather use Ingold's notion of what the landscape *is* to inspire a design game, it still requires that we actively work against thinking of the environment as a resource to be managed, and instead as an opportunity to think in terms of relation and process to collaboratively imagine new kinds of futures. The following sections use passages from Ingold to motivate various design choices and activities for another form of landscape game.

PLACES OF THE LANDSCAPE

"Spatial differentiation implies spatial segmentation. This is not so of the landscape, however. For a place in the landscape is not 'cut out' from the whole, either on the plane of ideas or on that of material substance. Rather, each place embodies the whole at a particular nexus within it, and in this respect is different from every other. A place owes its character to the experiences it affords to those who spend time there – to the sights, sounds and indeed smells that constitute its specific ambience. And these, in turn, depend on the kinds of activities in which its inhabitants engage. It is from this relational context of people's engagement with the world, in the business of dwelling, that each place draws its unique significance. Thus whereas with space, meanings are attached to the world, with the landscape they are gathered from it." (ibid., p.192)

In our design game, participants choose a starting point among different topographies, for example a gameboard with or without sea, coastline or forests, but still without settlements (Figure 5, left). To make the landscape more iconic and interpretable, we use historically inspired cartography for the gameboards and are not too concerned with ideas of scale or reference to actual locations (Figure 6). They then compose a landscape using game pieces that represent their future farms and—if so desired—other settlements, fixing them to the game-boards (Figure 5, center). Their motivation behind, for example, putting a farm close to the sea and far from a metropolitan city are shared with other players. Participants are asked to choose illustrations from a set of H.C. Andersen stories to represent their farms, integrating them into their descriptions, (Figure 5, right; Figure 7) as well as giving their farms a name. Follow-up questions into the sights, sounds and smells are meant to give their settings a particular ambience.

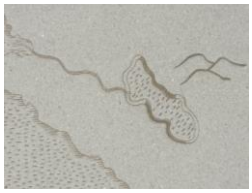


Figure 6: Gameboard map prototype with hills, a lake, and a stream near the coast (image by M.A. Padalak).

Leaving out details in the gameboard maps gives room for participants to fill in the blanks, and the cartographic materials' historical/fantasy aesthetics further pulls them away from the rationalist thinking most often tied to agricultural production—along with worries about the exact geographical locations of their landscapes. Contemporary maps usually only show a section of a whole and are often saturated by practical information and other 'real-world' connotations, rather than supporting open-ended imagination to elicit stories. Using ambiguous illustrations, that don't directly relate to farming, as basis for the stories about their farms is meant to further expand their imaginative thinking.

The goal of the first phase of our landscape game is to spark participants' imagination and have them consider the kind of place they would like to dwell in the future. Having participants compose and place the settlements is to support their reflection on what the places of their future farms afford them and those around them, also in terms of their relationality as inhabitants of these landscapes. Ingold believes that places in the landscape are defined by the experiences and activities of those who dwell there and can be neither separated nor isolated from them. They are defined by that nexus. As participants build maps of their own farms in the game board, they tell stories of how city and forest, farm, river, and sea are interrelated. The participants' stories are meaningful in themselves and across boundaries.



Figure 7: Example H.C Andersen illustration (image in the public domain Hansen, 1870).

WORK OF THE TASKSCAPE

"*I shall adopt the term 'task', defined as any practical operation, carried out by a skilled agent in an environment, as part of his or her normal business of life. In other words, tasks are the constitutive acts of dwelling. No more than features of the landscape, however, are tasks suspended in a vacuum. Every task takes its meaning from its position within an ensemble of tasks, performed in series or in parallel, and usually by many people working together. One of the great mistakes of recent anthropology…. has been to insist upon a separation between the domains of technical and social activity, a separation that has blinded us to the fact that one of the outstanding features of human technical practices lies in their embeddedness in the current of sociality. It is to the entire ensemble of tasks, in their mutual interlocking, that I refer by the concept of taskscape.*" (ibid., p. 195)

Performing everyday work routines or *tasks*, e.g. farm labor, makes participants landscape *dwellers*. The tasks' significance comes from their relation to one another and place in "the current of sociality", together, they are the *taskscape*. To expand the taskscape in our landscape game, participants list and describe tasks performed in their landscapes and the people—if any—that perform them.

Participants whose agricultural products require further processing, like dairy, slaughter or cooking, add the appropriate facilities to gameboards and connect them to their farms (Figure 8, left). Participants are then asked to consider which technologies, projected to have the biggest impact to their sector, they might see themselves implementing (Figure 8, right).

The increased efficiency and consolidation of farms over decades has shrunk the number of farmers, along with the demand for farmhands. Future technological development is likely to increase automation further, following a trend that has already changed the kind of sociality that once was inherent to farming, like calling in help for the harvest or simply having a person to milk cows—that now tends to be done by robots.

Asking what taskscape participants desire seems central: who (or what) performs tasks? Is it done locally or out of sight? What infrastructures will it require? How does future change affect farmers' stories and own place in the landscape? Historically, industry R&D has been most influential in imagining future agricultural practices. In this game, therefore, participants should be asked about the role they see for contemporary technologies (e.g. IoT, GIS, AI, automation, biotech, and the like) that the sector is investing in to get an initial understanding of how the participants envision farming is done.

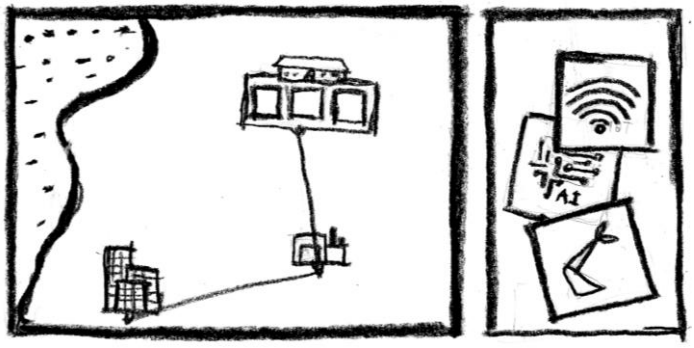


Figure 8: Two sketches of suggested activities and game cards. Left) A participant wanting to raise cows adds a dairy to the landscape and a line that traces the milk's journey from farm to dairy to consumer. Right) After participants have considered what infrastructure and machinery is needed for their productions, they are asked what future technologies, projected as most impactful, they see themselves adopt, like IoT, A.I., and automation (image by M.A. Padalak).

TEMPORALIZING THE LANDSCAPE (GAME)

"The world itself takes on the character of an organism, and the movements of animals—including those of us human beings—are parts or aspects of its life-process...This means that in dwelling in the world, we do not act upon it, or do things to *it; rather we move along* with *it. Our actions do not transform the world, they are part and parcel of the world's transforming itself. And that is just another way of saying that they belong to time."* (ibid., p. 200)

"Meaning is there to be discovered in the landscape, if only we know how to attend to it. Every feature, then, is a potential clue, a key to meaning rather than a vehicle for carrying it. This discovery procedure, wherein objects in the landscape become clues to meaning, is what distinguishes the perspective of dwelling. And since, as I have shown, the process of dwelling is fundamentally temporal, the apprehension of the landscape in the dwelling perspective must begin from a recognition of its temporality." (ibid., p. 208)

Participants are faced with potential future demands and events (Figure 9, left) that could be consequential for agricultural production (floods, legislation, changes in climate). They are asked to discuss, respond and adapt the landscapes accordingly (Figure 9, right). Infrastructure(s) might have to be re-built, moved or merged, technology and machinery adopted or retired, new responsibilities shared or divided. Whereas the preceding activities are mainly done individually, apart from sharing thoughts in plenary, these temporalizations are meant for the participants to collaborate together and decide on responses, reflecting their interconnectedness and co-dependence. Together they create the stories that we both inscribe into and read from the landscape, forming and enacting new forms of futures.

Like the world, the landscape is always changing, so to think that the landscape can be prepared for "when" the future arrives ignores that the future is continually

unfolding from the present. Components of the landscape—both animate and inanimate—together cocreate the landscape and to consider one in isolation from the rest, omits that their meaningfulness lies in their relationality. From the perspective of dwelling in a landscape, neither the landscape nor its components can be meaningfully considered without recognizing that they are temporal. Dwelling in the landscape over time has fundamental implications for how we imagine and tell stories of the future—it is never complete or finished, but always in flux.

Correspondingly, the stories of the farming landscape as inflected by technologies are, like the landscape, not considered "final" at any point. After participants' images of their farms have been narrated with stories, relations and hopes they should also be "traversed through time" to imagine new ways of being, to project impacts and effects, and to imagine different forms and means of dwelling together. This means to act out or anticipate what the future could bring to "temporalize" the stories being told. We posit that this can also make space to explore what happens when stories based in this emergent cocreated landscape collide with contested topics of public discourse.

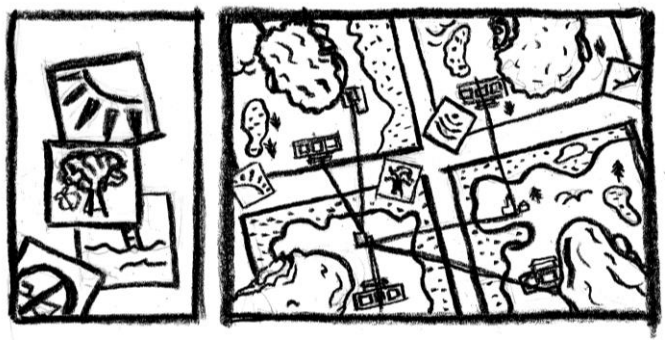


Figure 9: Two sketches of suggested activities and game cards. Left) Event-cards use recent and projected examples of obstacles to farmers, like solar storms interfering with GPS, political demand of more protected nature, rising sea-levels or disruption of global shipping. Right) Once stories have been told about the landscape in these stories, participants are tasked with discussing and responding to event-cards. This includes appropriately adapting and changing landscapes, dividing responsibilities, sharing infrastructure, adopting or rejecting technology, etc (image by M.A. Padalak).

DISCUSSION

The above approach iterates on an already-popular design game, to support a condensed iterative process of eliciting farm student's expectations of professional life to develop new stories that imagine new agricultural futures. The suggested activities aim to ground and give form to new future farming stories while relating them to other components of the landscape. This provides structure to expand them into new forms of agricultural imaginaries by posing questions inspired by Ingold to understand the make-up of these worlds, the practices that constitute a place and how these practices shape and are shaped by it. We don't intend for these participants to understand the landscape in new ways based on Ingold, but rather that his framing of the landscape can offer a means of eliciting stories about the components of landscapes that they are already familiar with. Multiple participants linking their own desired place to others' works to explore how future stories are entangled with imagined infrastructures in the world. At the same time, presenting participants with features and technologies from existing technical narratives explores how these technologies fit into future dwelling—if they fit at all.

MATERIALIZING POLITICIZED LANDSCAPES

The components of the design game—mapping a common farming landscape, placing tasks into the landscape as dwellers, and projecting how temporal shifts might affect dwelling—together serve to make the territory of farming contestable. This landscape game is a means to take a political issue into conversation via design games, offering scaffolding for participation in publics that might not be accounted for in terms of political conversations (DiSalvo et al., 2014).

Eriksen et al. have suggested design games can be used to explore power relations of people and materials, though mainly among those present in specific design game sessions (2014). Using Ingold's notion of landscape as a guide offers a way to tell concrete stories of farming not in the abstract, but as a practice that is situated and relational. Farming students playing the landscape game are invited to tell stories about farming futures that are as much about articulating present day issues as they are about the future as such. We hope that stories like these can support eliciting more robust imaginaries for future farming that can participate more actively in policymaking.

Political negotiation seeks quantifiable compromises, measures to determine what the industry and sector stakeholders may or may not do. Following Ingold, neither the landscape nor the taskscape can be asked how much of if there is, but only what it is like (2002). Our landscape game attempts to occlude these measurable quantities to develop broader and more aspirational futures than might be politically permissible. At the same time, tensions arise among farming students when the political and industry realities that they must face are brought together with the concerns of the dwellers that they will be as they co-create the landscape that they will be part of in the future.

DWELLING IN SPECULATIVE INFRASTRUCTURE

Part of what motivates this paper is the rallying cry that there is a crisis of imagination in agriculture futures. Put differently, the agrilogistic frame that contemporary agricultural practices operate inside of is conceptually totalizing. In *Capitalist Realism: Is There no Alternative*

Mark Fisher describes “the widespread sense that not only is capitalism the only viable political and economic system, but also that it is now impossible even to imagine a coherent alternative to it” (2009, p. 8). We see a similar issue with what we might call agricultural realism in our domain—"pragmatic” approaches to the issue foreclose any concepts that might lead to real change.

The idea of landscape, as above, makes the practice of farming already somewhat irrational, and hopefully can lead to new conceptualizations of a practice that seems fully rationalized. To subvert and explore new lifeworlds for farming, this design game operates in conversation with Wong et al.’s idea of “infrastructural speculation” (2020) that examines the lifeworlds of speculative design artifacts and their infrastructures by telling stories of dwelling in landscapes that are created by human and non-human relations. By focusing on practice and infrastructures, the materials of dwelling, we can imagine “staying with the trouble” (Haraway, 2016) of agriculture and have participants together explore the “complex middle ground between three supposed binaries: dystopia/utopia, working/broken, and future/past” (Wong et al., 2020, p. 9).

Other methods of producing new kinds of fables or stories for futures like fabulation (Søndergaard et al., 2023, Jenkins et al., 2023) operate in this way, and would work well in concert to develop more fully fleshed out narratives of future farming, reorienting these stories away from simplistic techno-fixes. The concept of the landscape works to site this fabulation in a world that has close ties to the material of farming—the land-based practices that make up its taskscape—and engage specifically with the environment. Aesthetically, the landscape offers a way to engage with environmental discourses as something that is practiced, intentional, and interrelated, and not merely as an image. After Svabo and Ekelund, “Environmental aesthetics need to develop a discourse that unfolds the natural (and cultural) as interrelations and not only as panoramas” (2015, p. 5). Rather than asking participants to inhabit, adjust or relate to fictional worlds or artefacts, the first order of business should be to attune to what and how they perceive and move through the landscapes that they know and want to dwell in themselves.

LANDSCAPES OF THE FUTURE

As formalized into a landscape game, Ingold’s notion of the landscape offers an interlinking of activity and place through the idea of dwelling, creating a fundamentally embedded and relational way of conceptualizing futures for design. Of course, design games already do much of this. However, we see Ingold’s notion as making us take the idea of landscape seriously, asking us to think explicitly about the relations that envisioning, occupying, and acting in a landscape are always already producing. We certainly don’t intend that this singular reference becomes dominant in this space, or to provide an exclusive, singular lens to understand what the landscape game can be in design research. Simply, we take the landscape game and interpret it using the landscape to create a situated and relational design game for farming, a political space that is complicated and holds many different stakeholders. The vision of farming as portrayed in game is fully romanticized, drawing on normative visions of farming’s past to elicit new stories of farming’s future—visions that may not resonate with all participants.

CONCLUSION

With respect to agricultural futures, the crisis of imagination referred to above demands new forms of engagement with those that are often left out of the frame—farmers—and correspondingly, new forms for developing future farming imaginaries. Ingold’s “The Temporality of the Landscape” gives a perspective on the landscape that is at odds with how the idea future farming tends to be envisioned, offering inspiration for design games to help support an underrepresented political public participate in a contested political topic.

This approach revolves around co-developing future imaginaries from farm students in design game landscapes, first by creating a *landscape* to set the grounds for imagination and then developing a shared *taskscape* to place and develop new forms of relations for farming practices. Designing this landscape game offers a means of both exploring new forms, relations, and features for future farming while keeping it rooted in ties to the past, both materially and methodologically. At the same time, we do not argue that these features are lacking in existing landscape games, however in this one they are explicitly accounted for in thinking through what the landscape is and is for.

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